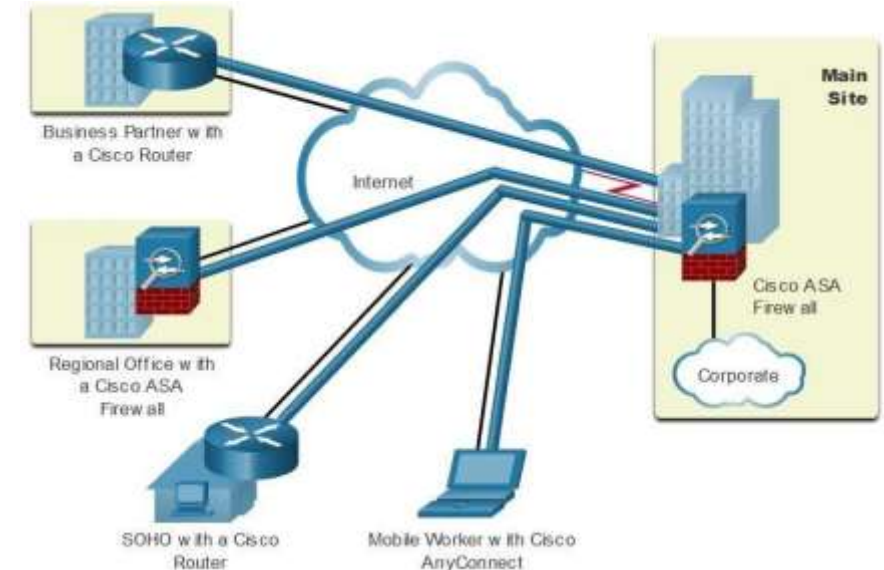


WAN CONCEPTS

Virtual Private Network (VPN)

VPN Technology

- To secure network traffic between sites and users, organizations use virtual private networks (VPNs) to create end-to-end private network connections.
 - A VPN is *virtual* in that it carries information within a private network, but that information is transported over a public network.
 - A VPN is *private* in that the traffic is encrypted to keep the data confidential while it is transported across the public network.
- The figure shows a collection of various types of VPNs managed by an enterprise's main site. The tunnel enables remote sites and users to access main site's network resources securely.



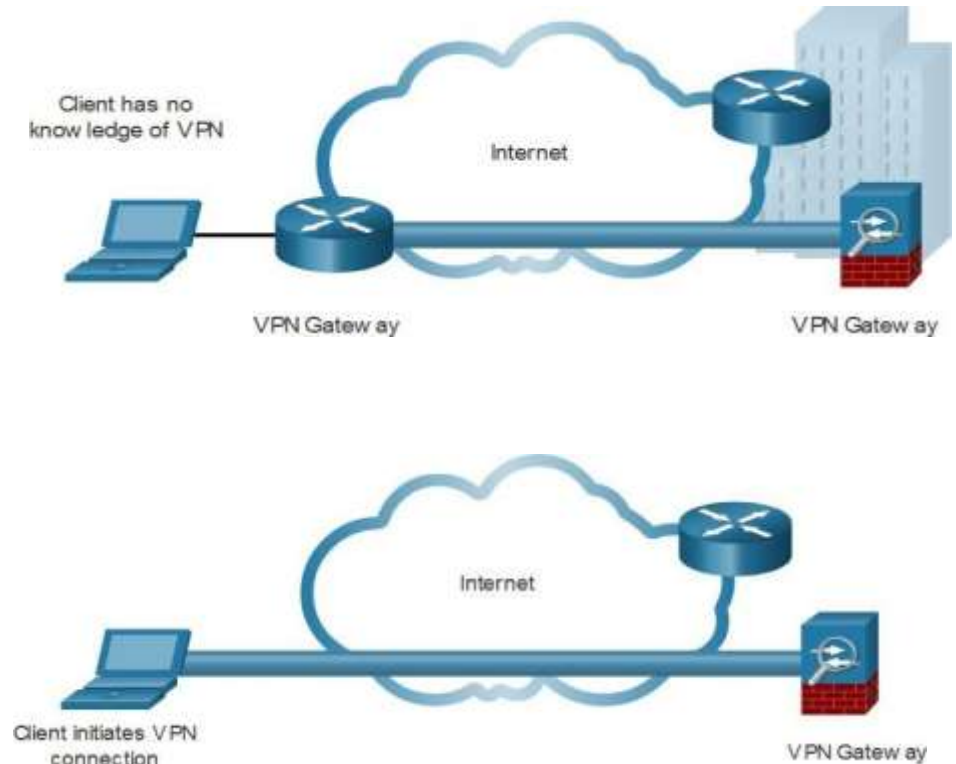
Benefits of VPN

- Modern VPNs now support encryption features, such as **Internet Protocol Security (IPsec)** and **Secure Sockets Layer (SSL)** VPNs to secure network traffic between sites.
- Major benefits of VPNs are shown in the table.

Benefit	Description
Cost Savings	With the advent of cost-effective, high-bandwidth technologies, organizations can use VPNs to reduce their connectivity costs while simultaneously increasing remote connection bandwidth.
Security	VPNs provide the highest level of security available, by using advanced encryption and authentication protocols that protect data from unauthorized access.
Scalability	VPNs allow organizations to use the internet, making it easy to add new users without adding significant infrastructure.
Compatibility	VPNs can be implemented across a wide variety of WAN link options including all the popular broadband technologies. Remote workers can take advantage of these high-speed connections to gain secure access to their corporate networks.

Site-to-Site and Remote Access VPNs

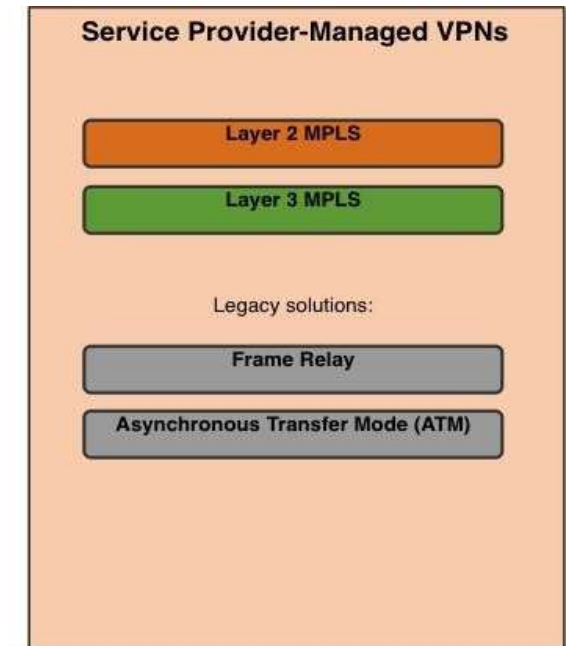
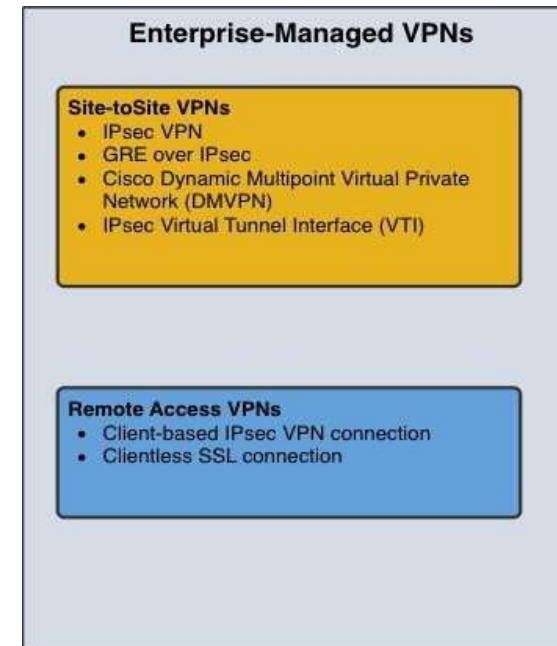
- A **site-to-site VPN** is created when VPN terminating devices, also called **VPN gateways**, are preconfigured with information to establish a secure tunnel. VPN traffic is only encrypted between these devices. Internal hosts have no knowledge that a VPN is being used.
- A **remote-access VPN** is dynamically created to establish a secure connection between a client and a VPN terminating device. For example, a remote access SSL VPN is used when you check your banking information online.



Enterprise and Service Provider VPNs

VPNs can be managed and deployed as:

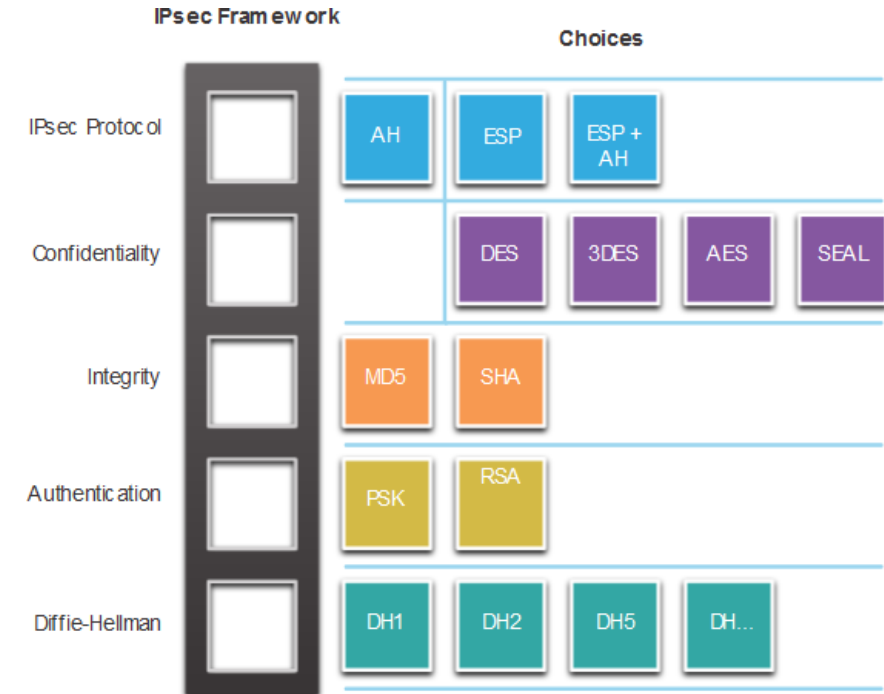
- **Enterprise VPNs** - common solution for securing enterprise traffic across the internet. Site-to-site and remote access VPNs are created and managed by the enterprise using IPsec and SSL VPNs.
- **Service Provider VPNs** - created and managed by the provider network. The provider uses Multiprotocol Label Switching (MPLS) at Layer 2 or Layer 3 to create secure channels between an enterprise's sites, effectively segregating the traffic from other customer traffic.



IPsec

IPsec is an IETF standard that defines how a VPN can be secured across IP networks. IPsec protects and authenticates IP packets between source and destination and provides these essential security functions:

- **Confidentiality** - Uses encryption algorithms to prevent cybercriminals from reading the packet contents.
- **Integrity** - Uses hashing algorithms to ensure that packets have not been altered between source and destination.
- **Origin authentication** - Uses the Internet Key Exchange (IKE) protocol to authenticate source and destination.
- **Diffie-Hellman** - Used to secure key exchange.

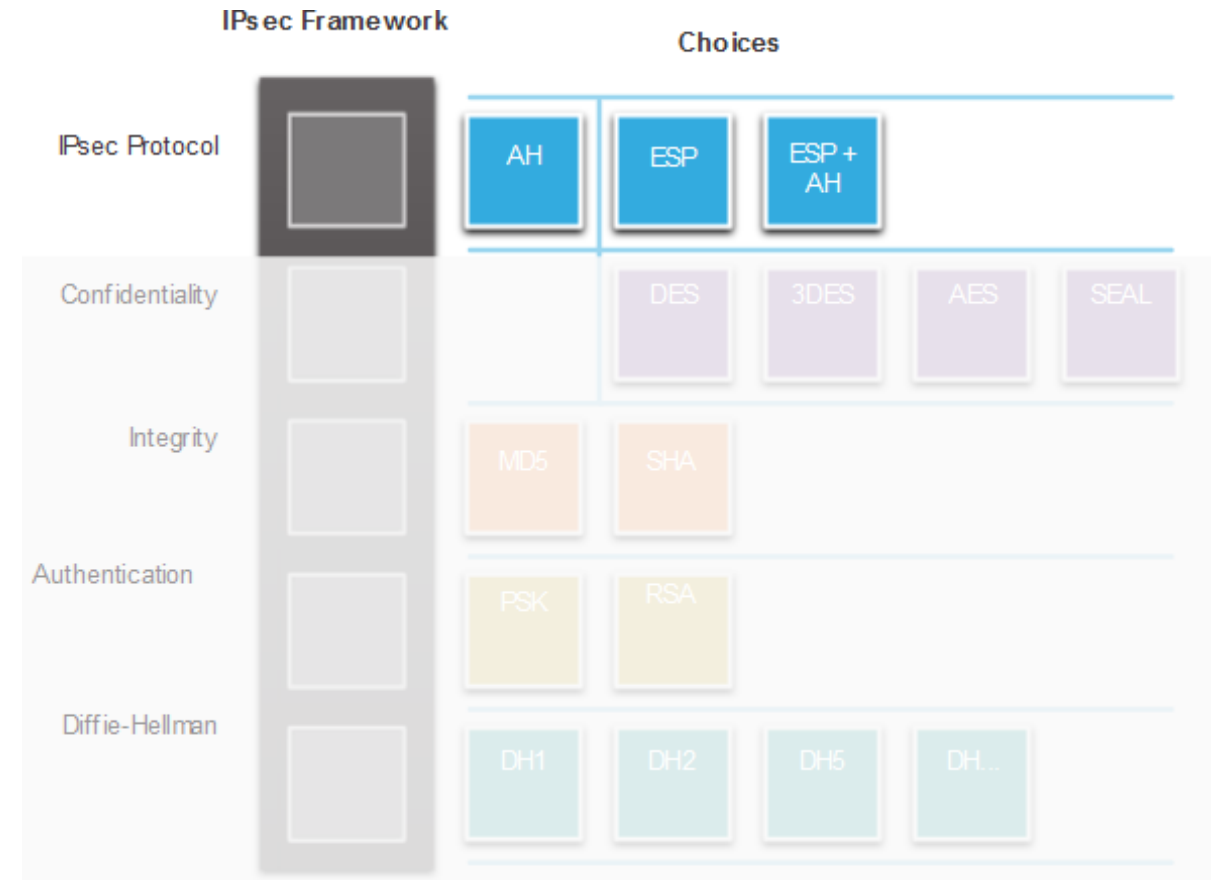


IPsec Security Functions

IPsec Function	Description
IPsec Protocol	The choices for IPsec Protocol include Authentication Header (AH) or Encapsulation Security Protocol (ESP). AH authenticates the Layer 3 packet. ESP encrypts the Layer 3 packet. Note: ESP+AH is rarely used as this combination will not successfully traverse a NAT device.
Confidentiality	Encryption ensures confidentiality of the Layer 3 packet. Choices include Data Encryption Standard (DES), Triple DES (3DES), Advanced Encryption Standard (AES), or Software-Optimized Encryption Algorithm (SEAL).
Integrity	Ensures that data arrives unchanged at the destination using a hash algorithm, such as message-digest 5 (MD5) or Secure Hash Algorithm (SHA).
Authentication	IPsec uses Internet Key Exchange (IKE) to authenticate users and devices that can carry out communication independently. IKE uses several types of authentication, including username and password, one-time password, biometrics, pre-shared keys (PSKs), and digital certificates using the Rivest, Shamir, and Adleman (RSA) algorithm.
Diffie-Hellman	IPsec uses the DH algorithm to provide a public key exchange method for two peers to establish a shared secret key. There are several different groups to choose from including DH14, 15, 16 and DH 19, 20, 21 and 24. DH1, 2 and 5 are no longer recommended.

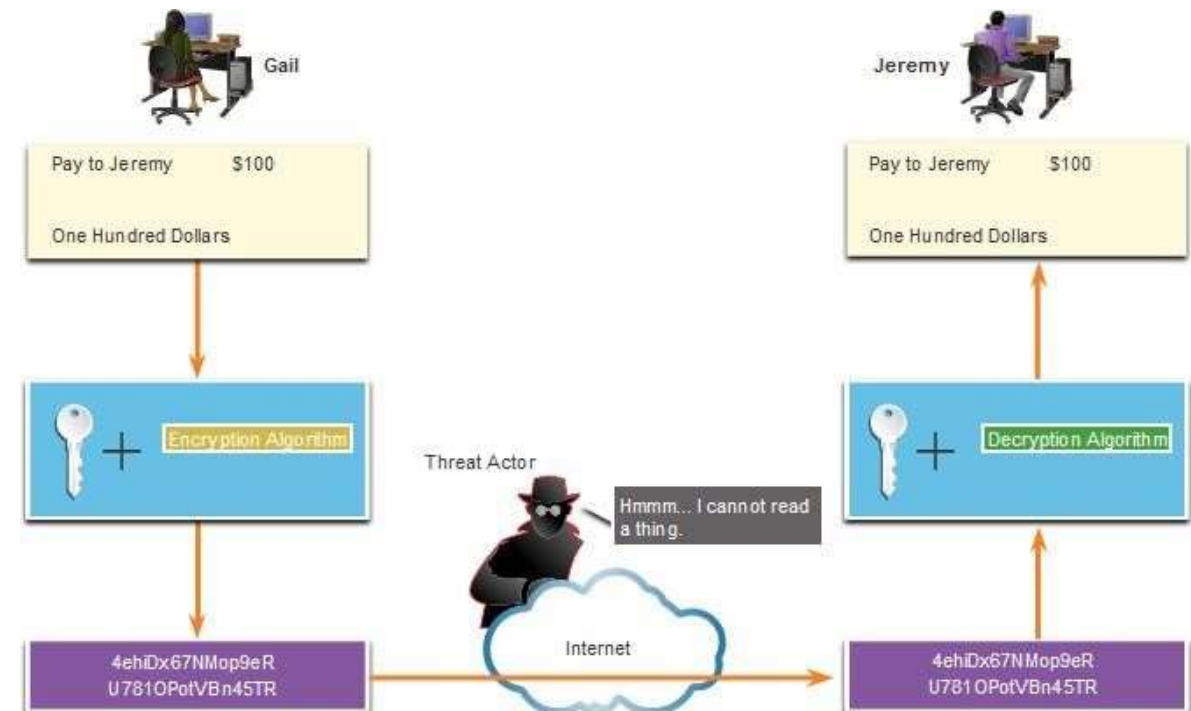
IPsec Functions: IPsec Protocol Encapsulation

- Choosing the IPsec protocol encapsulation is the first building block of the framework.
- Ipsec encapsulates packets using Authentication Header (AH) or Encapsulation Security Protocol (ESP).
- The choice of AH or ESP establishes which other building blocks are available.
- AH is appropriate only when confidentiality is not required or permitted.
- ESP provides both confidentiality and authentication.



IPsec Functions: Confidentiality

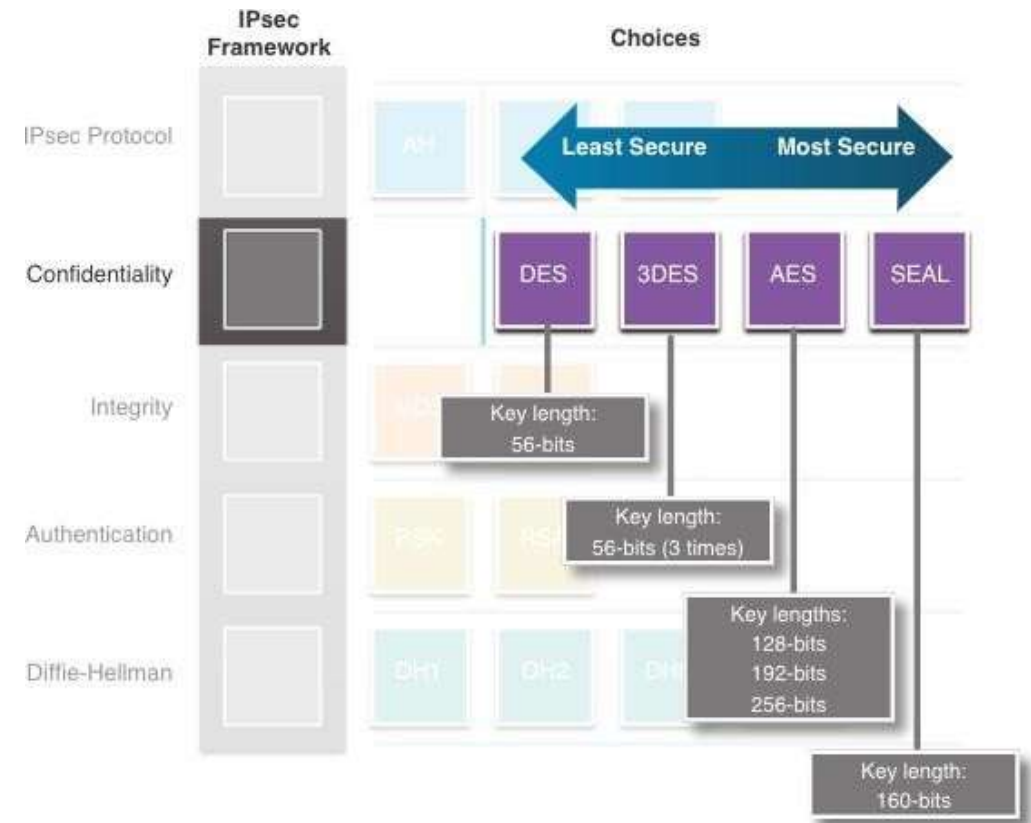
- The degree of confidentiality depends on the encryption algorithm and the length of the key used in the encryption algorithm.
- The number of possibilities to try to hack the key is a function of the length of the key - the shorter the key, the easier it is to break.



IPsec Functions: Confidentiality (Contd.)

The encryption algorithms highlighted in the figure are all **symmetric key cryptosystems**:

- **DES** uses a **56-bit** key.
- **3DES** uses **three independent 56-bit encryption keys per 64-bit block**.
- **AES** offers **three different key lengths: 128 bits, 192 bits, and 256 bits**.
- **SEAL** is **a stream cipher**, which means it **encrypts data continuously rather than encrypting blocks of data**. SEAL uses a **160-bit** key.



IPsec Functions: Integrity

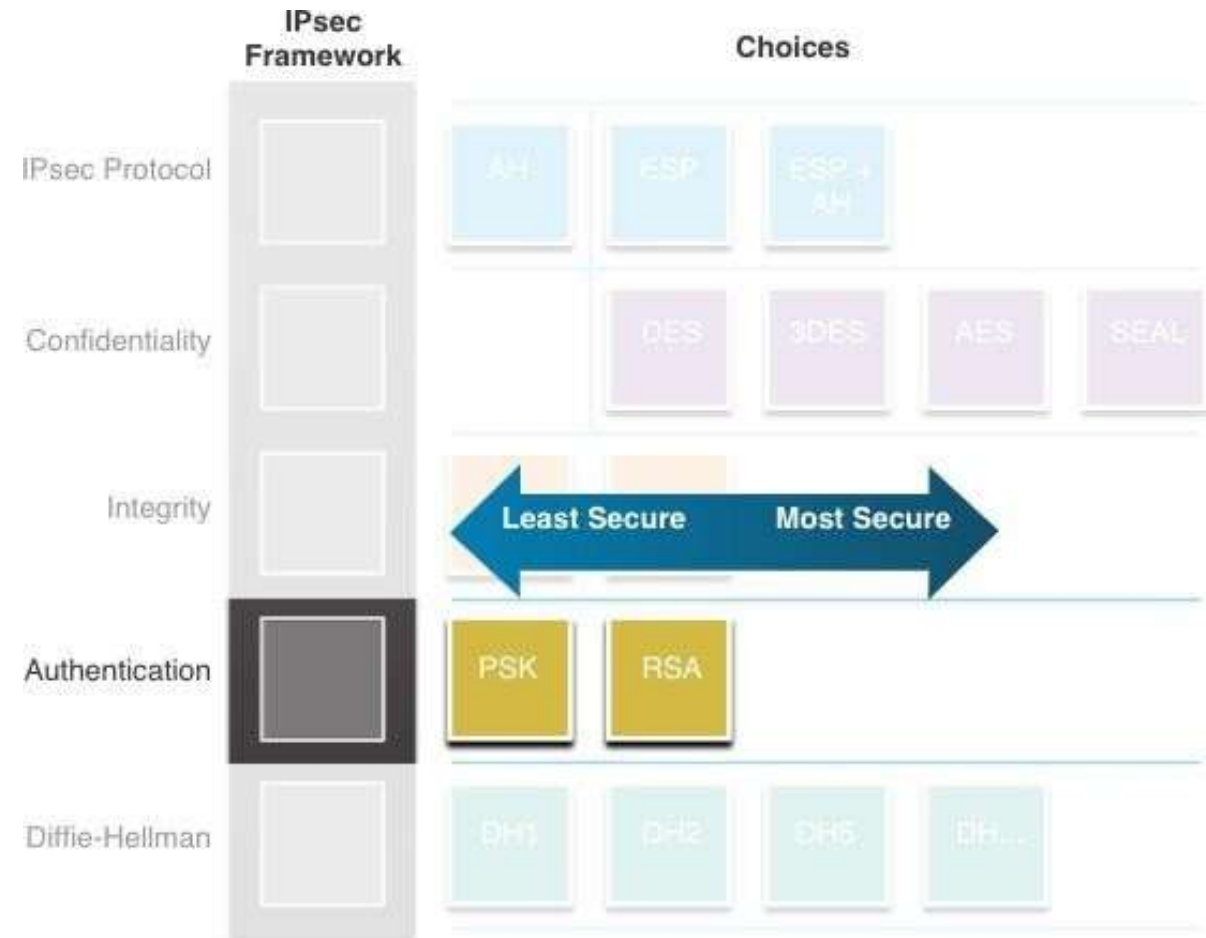
- **Data integrity** means that the data has not changed in transit.
- A method of proving data integrity is required.
- The **Hashed Message Authentication Code (HMAC)** is a data integrity algorithm that guarantees the integrity of the message using a hash value.
- **Message-Digest 5 (MD5)** uses a 128-bit shared-secret key.
- The **Secure Hash Algorithm (SHA)** uses a 160-bit secret key.



IPsec Functions: Authentication

There are **two IPsec peer authentication methods**:

1. **Pre-shared key (PSK)** - (PSK) value is **entered into each peer manually**.
 - Easy to configure manually
 - Does not scale well
 - Must be configured on every peer
2. **Rivest, Shamir, and Adleman (RSA)** - authentication uses **digital certificates** to authenticate the peers.
 - Each peer must authenticate its opposite peer before the tunnel is considered secure.

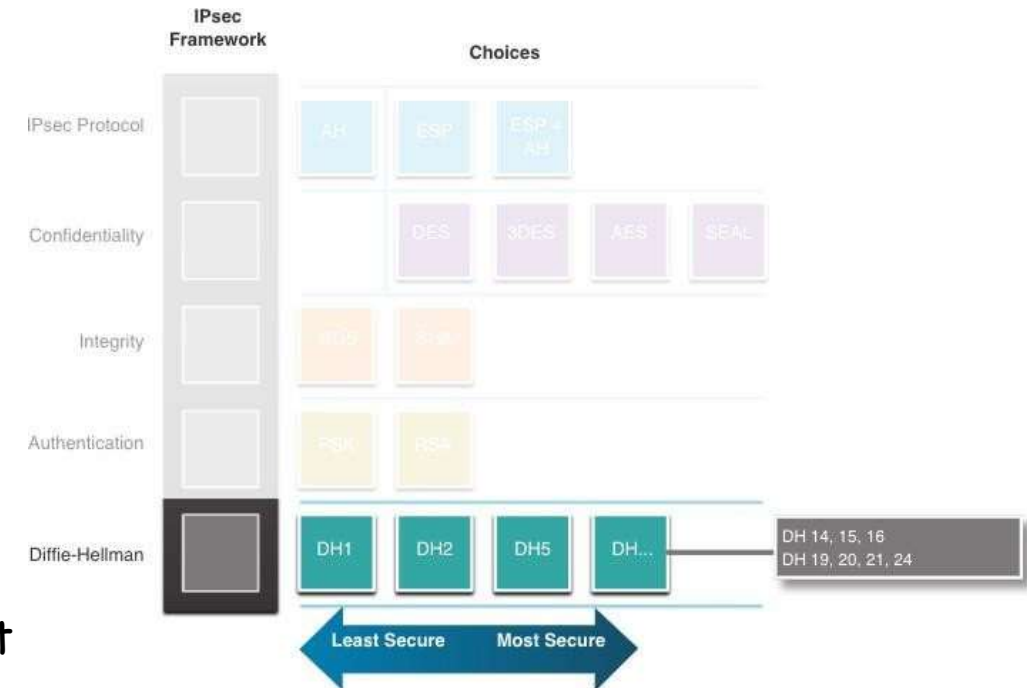


IPsec Functions: Secure Key Exchange with Diffie - Hellman

DH provides allows two peers to establish a shared secret key over an insecure channel.

Variations of the DH key exchange are specified as DH groups:

- DH groups 1, 2, and 5 should no longer be used.
- DH groups 14, 15, and 16 use larger key sizes with 2048 bits, 3072 bits, and 4096 bits, respectively
- DH groups 19, 20, 21 and 24 with respective key sizes of 256 bits, 384 bits, 521 bits, and 2048 bits support Elliptical Curve Cryptography (ECC), which reduces the time needed to generate keys.



IPsec Transport and Tunnel Mode

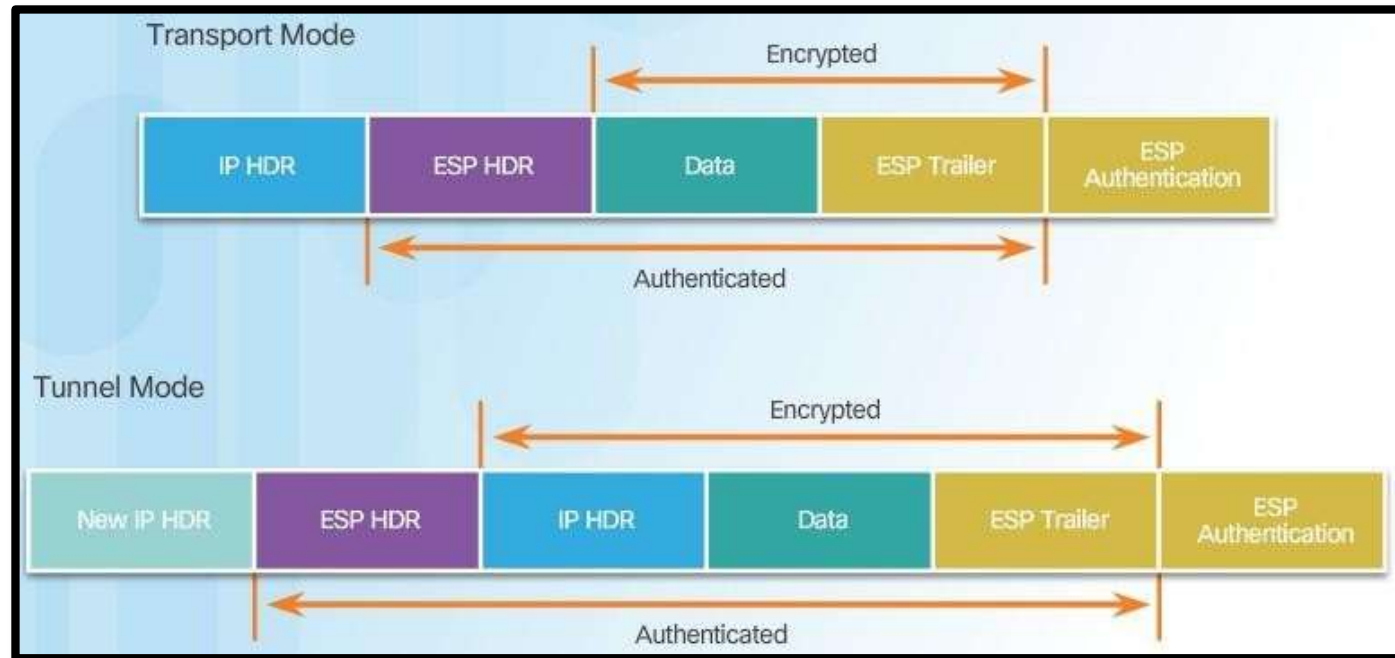
Transport and Tunnel Modes

- **Transport Mode** - In transport mode, security is provided only for the transport layer of the OSI model and above. Transport mode protects the payload of the packet but leaves the original IP address in plaintext. The original IP address is used to route the packet through the internet. ESP transport mode is used between hosts.
- **Tunnel Mode** - Tunnel mode provides security for the complete original IP packet. The original IP packet is encrypted and then it is encapsulated in another IP packet. This is known as IP-in-IP encryption. The IP address on the outside IP packet is used to route the packet through the internet.

Apply ESP in Transport and Tunnel Modes

Transport and Tunnel Modes

Apply ESP in Two Modes



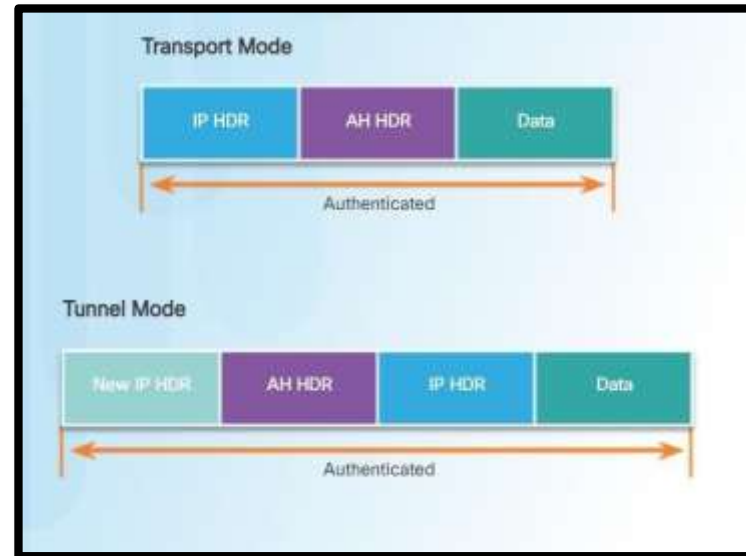
ESP transport mode is used between hosts.

ESP tunnel mode is used between a host and a security gateway, or between two security gateways

Apply AH in Transport and Tunnel Modes

Transport and Tunnel Modes

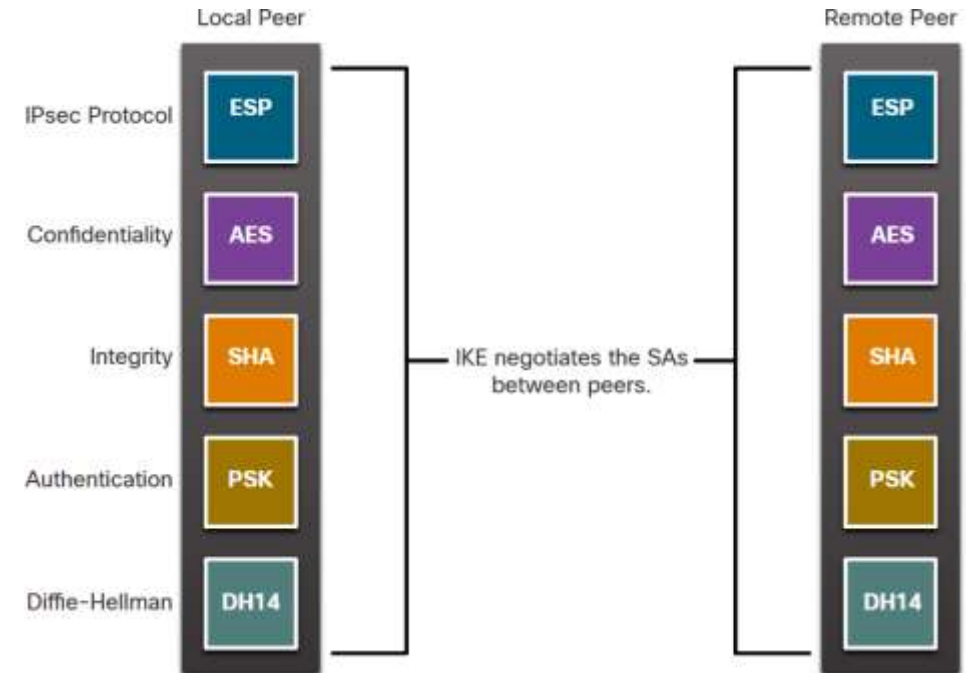
Apply AH in Two Modes



- AH transport mode provides authentication and integrity for the entire packet.
- It does not encrypt the data, but it is protected from modification.
- AH tunnel mode encapsulates the IP packet with an AH and a new IP header and signs the entire packet for integrity and authentication.

Internet Key Exchange (IKE)

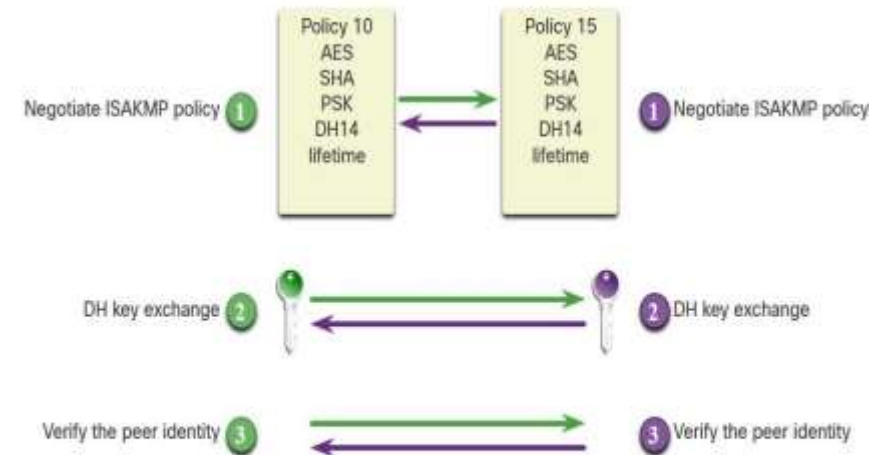
- IKE enhances Isec by adding features and simplifies configuration for the IPsec standard.
- IKE is a hybrid protocol that implements key exchange protocols inside the Internet Security Association Key Management Protocol (ISAKMP) framework.
- Instead of transmitting keys directly across a network, IKE calculates shared keys based on the exchange of a series of data packets.
- UDP port 500 packets must be permitted on any IP interface that is connecting a security gateway peer.



Internet Key Exchange (IKE) (Contd.)

- IKE uses ISAKMP for phase 1 and phase 2 of key negotiation.
- Phase 1 negotiates a security association (a key) between two IKE peers. The key negotiated in phase 1 enables IKE peers to communicate securely in phase 2. During phase 2 negotiation, IKE establishes keys (security associations) for other applications, such as IPsec.
- In Phase 1, two IPsec peers perform the initial negotiation of SAs. The basic purpose of Phase 1 is to negotiate ISAKMP policy, authenticate the peers, and set up a secure tunnel between the peers. This tunnel will then be used in Phase 2 to negotiate the IPsec policy.

Phase 1 - Negotiate ISAKMP policy to create a tunnel.



Phase 2 - Negotiate IPsec policy for sending secure traffic across the tunnel.



Internet Key Exchange (IKE) (Contd.)

- IKE Phase 2 is called quick mode and can only occur after IKE has established a secure tunnel in Phase 1. Quick mode negotiates the IKE Phase 2 SAs.
- In this phase, the SAs that IPsec uses are unidirectional; therefore, a separate key exchange is required for each data flow.



Internet Key Exchange (IKE) (Contd.)

IPsec Negotiation

IPsec negotiation to establish a VPN involves five steps, which include IKE Phase 1 and Phase 2:

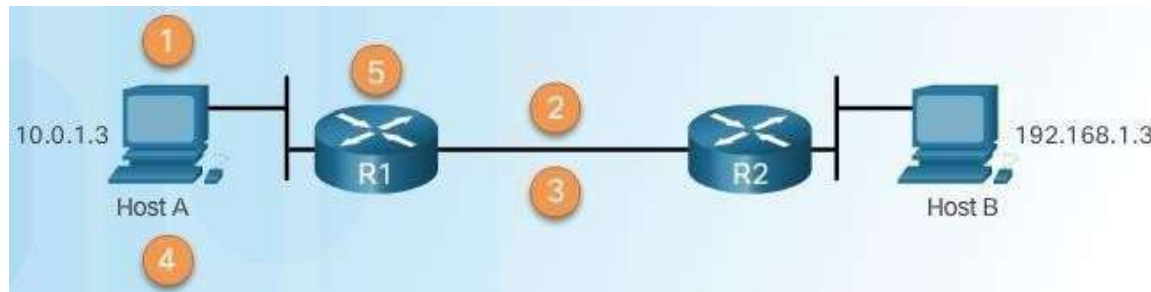
1. An ISAKMP tunnel is initiated when host A sends "interesting" traffic to host B. Traffic is considered interesting when it travels between the peers and meets the criteria that are defined in an ACL.
2. IKE Phase 1 begins. The peers negotiate the ISAKMP SA policy. When the peers agree on the policy and are authenticated, a secure tunnel is created.
3. IKE Phase 2 begins. The IPsec peers use the authenticated secure tunnel to negotiate the IPsec SA policy.

The negotiation of the shared policy determines how the IPsec tunnel is established.

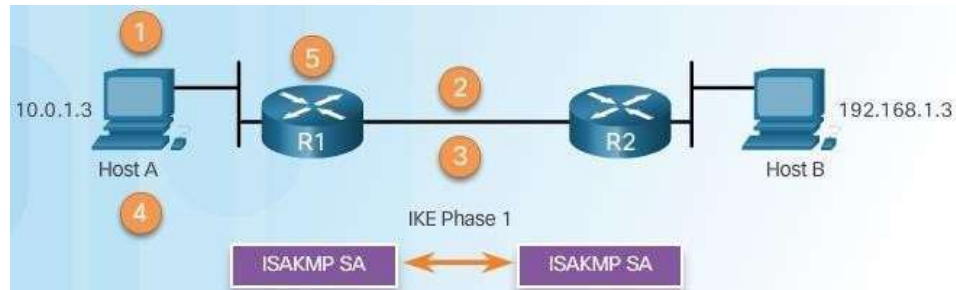
4. The IPsec tunnel is created, and data is transferred between the IPsec peers based on the IPsec SAs.
5. The IPsec tunnel terminates when the IPsec SAs are manually deleted, or when their lifetime expires.

Internet Key Exchange (IKE) (Contd.)

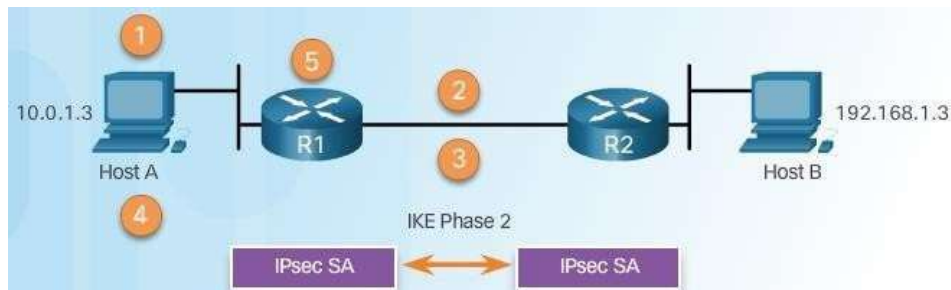
IPsec Negotiation



Step 1: Host A sends interesting traffic to Host B.



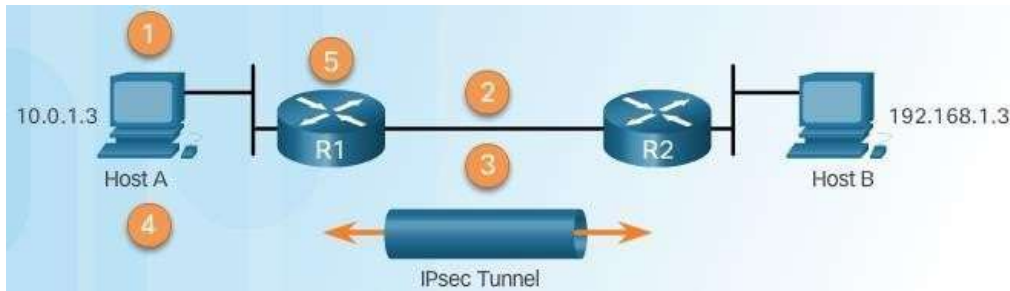
Step 2: R1 and R2 negotiate an IKE Phase 1 session.



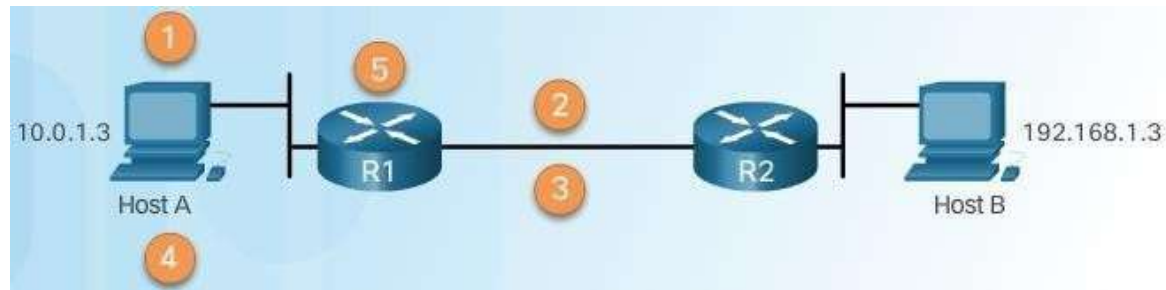
Step 3: R1 and R2 negotiate an IKE Phase 2 session.

Internet Key Exchange (IKE) (Contd.)

IPsec Negotiation



Step 4: Information is exchanged via IPsec tunnel.



Step 5: The IPsec tunnel is terminated.

Lab Activities-V

- Lab 5.1 Configure and Verify a Site-to-Site IPsec VPN